DARSHAN UNIVERSITY

Department of Computer Science & Engineering

Course Code: 2601CS102

Web Designing

B.Tech. First Semester

CHAPTER 1

HTML & Fundamentals of Web Designing

E-Notes / Study Material

Topic	Coverage
Internet & Web Architecture	Internet, WWW, Client-Server, HTTP Protocol
Web Technologies Overview	HTML, CSS, JavaScript, Bootstrap, Material UI
Web Design Principles	Effective Design, Navigation, Usability Issues
HTML Page Structure & Meta	DOCTYPE, head, body, meta tags
Text Formatting & Lists	Headings, paragraphs, formatting, lists, anchors
Tables & Forms	Table structure, form elements, input types
Multimedia & HTML5	Audio, video, iframe, semantic elements, validation

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1. Introduction to Internet, Web & Architecture

1.1 Internet vs. World Wide Web (WWW)

Although both terms are often used interchangeably in everyday conversation, the Internet and the World Wide Web are two very different things. Understanding this distinction is the first step in web designing.

Feature	The Internet	World Wide Web (WWW)
Definition	Global infrastructure of hardware & networks	Information system of software & web resources
Protocols Used	TCP/IP, SMTP, FTP, UDP, SSH	HTTP, HTTPS, WebSocket
Main Components	Computers, routers, servers, cables	Web pages, HTML files, media, URLs
Invented By	ARPA (US Govt.) — Late 1969s	Tim Berners-Lee at CERN — 1989
Example Services	Email, FTP, Online Gaming, VoIP	Websites, Web Apps, Online Shopping

💡 **Easy Way to Remember:**
Internet = The Roads & Highways (Physical infrastructure)
WWW = The Buildings, Shops & Offices on those roads (Services & Content)
You need the roads (Internet) to reach the buildings (Web)!

1.2 Client-Server Architecture

Web communication follows the Client-Server Architecture — a model where one device (client) requests services and another device (server) provides them.

How It Works — Step by Step:
Step 1 → User types a URL (e.g., www.google.com) in the browser
Step 2 → Browser sends an HTTP Request to the Web Server
Step 3 → Server processes the request and finds the required files
Step 4 → Server sends back an HTTP Response (HTML, CSS, Images)
Step 5 → Browser renders and displays the web page to the user

▸ Key Components:

- **Client (Web Browser):** The software that requests and displays web pages. Examples: Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge.
- **Web Server:** A powerful computer that stores websites and sends them to clients on request. Examples: Apache, Nginx, IIS.

- **URL (Uniform Resource Locator):** The web address used to locate a specific resource. Example: https://www.darshan.ac.in/index.html
- **IP Address:** A unique numerical label assigned to every device on the Internet (like a home address for computers).

1.3 HTTP Protocol (HyperText Transfer Protocol)

HTTP is an application-layer protocol that defines the rules for transferring data (web pages, images, videos) between a web client and a web server. It is the backbone of data communication on the World Wide Web.

Key Characteristics of HTTP:

Stateless — Each request is completely independent; the server has no memory of past requests from the same client

Request-Response Model — Client always sends a Request; Server always sends back a Response

Text-Based — HTTP messages are written in plain readable text format

HTTPS = HTTP + SSL/TLS Encryption — Adds a security layer to protect data in transit

▸ Common HTTP Request Methods:

Method	Full Form	Purpose	Example Use Case
GET	Get Data	Fetches/retrieves data from the server	Loading a webpage, searching Google
POST	Post Data	Sends data to the server (in request body)	Submitting a login form
PUT	Update Data	Replaces/updates an existing resource	Updating a user profile
DELETE	Delete Data	Removes a specified resource from server	Deleting an account
PATCH	Partial Update	Partially modifies an existing resource	Changing just the username

▸ Common HTTP Status Codes:

Code Range	Category	Common Examples
1xx	Informational	100 Continue — Request received, process continuing
2xx	Success ✓	200 OK — Request successful | 201 Created — Resource created
3xx	Redirection	301 Moved Permanently | 304 Not Modified (use cached version)
4xx	Client Error ✗	400 Bad Request | 401 Unauthorized | 404 Not Found (most common!)

5xx	Server Error ✗	500 Internal Server Error | 503 Service Unavailable

2. Overview of Web Technologies Stack

Modern web development is built on a layered stack of technologies, each serving a specific and distinct purpose. Think of it as building a house — you need a foundation, walls, paint, and furniture separately.

The Web Technology Stack (Bottom to Top):
Layer 4 — UI Frameworks (Bootstrap, Material UI) → Pre-built, reusable components
Layer 3 — JavaScript → Interactivity, dynamic behavior, API calls
Layer 2 — CSS / CSS3 → Visual styling, layout, animations, colors
Layer 1 — HTML / HTML5 → Document structure, content, and meaning

Technology Breakdown

Technology	Role	Core Responsibility	Real Example
HTML5	Structure	Defines content, headings, forms, media elements	<h1>, <p>, <img>, <form>
CSS3	Presentation	Styles colors, typography, spacing, layout, animations	background: blue; display: flex;
JavaScript	Behavior	Handles interactivity, form validation, dynamic updates	Button click events, API calls
Bootstrap 5	Responsive Framework	Pre-styled grid system, buttons, navbars, cards	class="col-md-6 btn btn-primary"
Material Design	Design System	Google's UI guidelines for consistency and visual quality	Elevation, FAB, ripple effects

💡 **Analogy for Students:**
HTML = Skeleton of the human body (structure, bones)
CSS = Skin, clothes, and appearance (how it looks)
JavaScript = Brain and muscles (thinking and movement)
Bootstrap = Ready-made clothes from a store (pre-built components)

3. Principles of Web Design & Navigation Architecture

3.1 Concepts of Effective Web Design

Good web design is not just about making a page look pretty. It is about making information easy to find, easy to read, and pleasant to use. Here are the core principles every web designer must follow:

Design Principle	What It Means	Practical Example
Visual Hierarchy	Arrange elements by order of importance using size, color, weight	Make the main heading larger than sub-headings
Simplicity & Clarity	Remove unnecessary clutter; use white space generously	Google's clean homepage — just a search bar
Consistency	Same fonts, colors, button styles on every page	Use only 2–3 font types across the website
F-Shape Reading Pattern	Design for how users naturally read (top-left first, then scan down)	Place important content at the top-left area
Contrast	Ensure text stands out clearly from its background	Black text on white background for readability
Mobile-First Design	Design for mobile screens first, then scale up to desktop	Responsive layouts using CSS media queries

3.2 Key Web Design Issues

Designers and developers must address these common challenges to ensure a website works well for all users:

- **Cross-Browser Compatibility:** Web pages must render correctly in Chrome (Blink engine), Firefox (Gecko engine), Safari (WebKit engine), and Edge. Different browsers interpret CSS differently.
- **Responsive Screen Resolution:** Pages must adapt to screen sizes from 320px (mobile phones) to 4K desktop monitors. CSS media queries and flexible grids solve this.
- **Page Load Speed:** Heavy uncompressed images, too many HTTP requests, and unminified code slow down pages. Target: under 3 seconds load time. Bounce rate increases 32% when load time goes from 1s to 3s.
- **Accessibility (WCAG Standards):** Websites must be usable by people with visual, motor, or cognitive disabilities. Use proper color contrast, alt text for images, keyboard navigation, and ARIA roles.
- **SEO (Search Engine Optimization):** Proper use of semantic HTML tags helps search engines like Google understand and rank your content correctly.

3.3 Designing Effective Navigation

Good navigation helps users find what they need in as few clicks as possible. If users cannot navigate easily, they leave the site.

Navigation Type	Description	Best Used For
Primary Navigation Bar	Main links at the top or left side of every page	Main sections: Home, About, Courses, Contact
Breadcrumb Trail	Shows user's current location: Home > Courses > HTML	Deep hierarchical websites, e-commerce sites
Footer Navigation	Links at the bottom of the page	Sitemap, Terms, Privacy Policy, Social links
Hamburger Menu (☰)	Collapsed menu icon for mobile screens	Mobile-responsive navigation
Sticky Navigation	Navigation bar stays fixed while scrolling down	Long pages, documentation sites

The Three-Click Rule:

Users should be able to find any piece of information on your website within 3 mouse clicks from the homepage.

If it takes more than 3 clicks → your navigation needs to be restructured and simplified.

4. HTML Page Architecture & Metadata

4.1 Basic Structure of an HTML5 Page

Every valid HTML5 document follows a specific, required structure. This is like the skeleton that holds the entire page together. You must include every part shown below:

```
<!DOCTYPE html>
<html lang="en">

  <head>
    <!-- Metadata section — NOT visible on the page -->
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My First Web Page</title>
    <link rel="stylesheet" href="style.css">
  </head>

  <body>
    <!-- Visible content goes here -->
    <h1>Hello, World!</h1>
    <p>Welcome to Web Designing — Chapter 1.</p>
  </body>

</html>
```

▸ Explanation of Each Part:

Tag / Element	Purpose & Meaning
<!DOCTYPE html>	Tells the browser: 'This is an HTML5 document.' Must be the very first line.
<html lang="en">	Root element that wraps everything. lang="en" specifies the language as English.
<head>	Container for metadata (information about the page). NOT displayed to the user.
<meta charset="UTF-8">	Supports all characters including Hindi, Chinese, symbols, and emojis.
<title>Page Title</title>	Text that appears on the browser tab (important for SEO too).
<link rel="stylesheet">	Links an external CSS file to apply styling to the page.
<body>	Contains ALL visible content: headings, text, images, tables, forms, videos.

4.2 Meta Tags (<meta>)

Meta tags provide information about a web page to browsers and search engines. They are placed inside the <head> section and are NOT visible to users on the page itself.

```
<!-- 1. Character Encoding — supports all languages & symbols -->
<meta charset="UTF-8">

<!-- 2. Responsive Mobile Display — essential for all modern websites -->
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<!-- 3. SEO — page description shown in Google search results -->
<meta name="description" content="Learn HTML and Web Designing basics.">

<!-- 4. SEO — keywords that help search engines categorize the page -->
<meta name="keywords" content="HTML, CSS, Web Design, B.Tech, Darshan University">

<!-- 5. Author information -->
<meta name="author" content="Dept. of Computer Science, Darshan University">

<!-- 6. Auto-refresh the page every 30 seconds -->
<meta http-equiv="refresh" content="30">
```

⚠️ **Important for Students:**

The viewport meta tag is MANDATORY for all modern responsive websites.

Without it, mobile phones will display the full desktop version shrunken down.

Always add: <meta name="viewport" content="width=device-width, initial-scale=1.0">

5. Text Elements, Formatting & Organization

5.1 Headings and Paragraphs

HTML provides 6 levels of headings (h1 to h6) and a paragraph tag. Headings define the hierarchy and structure of your content, similar to chapters and sub-chapters in a book.

```
<h1>Main Page Title – Largest (Only ONE per page)</h1>
<h2>Major Section Heading</h2>
<h3>Sub-Section Heading</h3>
<h4>Sub-Sub-Section</h4>
<h5>Minor Heading</h5>
<h6>Smallest Heading – Least Important</h6>

<p>This is a standard paragraph of body text.
   It wraps automatically to the next line.</p>

<p>Each paragraph creates a new block with automatic spacing.</p>
```

SEO Rule:
Always use only ONE <h1> per page — it represents the main topic of the page.
Use <h2> for major sections, <h3> for sub-sections, and so on in logical order.
Never skip heading levels (don't jump from h1 directly to h4).

5.2 Text Formatting Tags

HTML provides inline tags to add semantic meaning and visual emphasis to specific words or phrases within a paragraph:

Tag	Visual Output	Meaning / Purpose	Example Code
	Bold text	Visual bold — no semantic meaning	Important
	Bold text	Bold + semantically important (screen readers say it louder)	Warning!
	Italic text	Visual italic — no semantic meaning	Book Title
	Italic text	Italic + emphasized meaning (stress emphasis)	must read this
<mark>	Highlighted text	Highlights text like a yellow marker pen	<mark>key term</mark>
<small>	Smaller text	Fine print, copyright, side notes	© 2025
<del>	~~Strikethrough ~~	Represents deleted / removed content	~~Old Price~~

<ins>	Underlined text	Represents newly inserted content	<ins>New Price</ins>
<sub>	Below baseline	Subscript — for chemical formulas etc.	H₂O
<sup>	Above baseline	Superscript — for mathematical powers	E = mc²
<code>	Monospace code	Displays inline code snippets	<code>color: red;</code>

5.3 Container & Organization Elements

- ** (Division Block):** Block-level container. Starts on a new line and takes full width. Used to group and style sections of the page.
- ** (Inline Container):** Inline container. Does NOT break the line flow. Used to style a specific word or phrase within a paragraph.
- **
 (Line Break):** Creates a single line break. Self-closing tag (no </br> needed).
- **<hr> (Horizontal Rule):** Draws a horizontal dividing line across the page to separate content sections.
- **<blockquote>:** Used for long quotations from external sources. Displayed with indentation.
- **<pre> (Preformatted Text):** Preserves all spaces, tabs, and line breaks exactly as typed. Used for code displays.

6. HTML Lists, Anchors & Images

6.1 HTML Lists

HTML supports three types of lists to organize related items of information in a structured, readable format:

▸ A. Unordered List <ul> — Bullet Points

Used when the order of items does NOT matter (e.g., a list of features, ingredients, or options):

```
<ul>
  <li>HTML5 – Page Structure</li>
  <li>CSS3 – Page Styling</li>
  <li>JavaScript – Interactivity</li>
  <li>Bootstrap – Responsive Design</li>
</ul>
```

▸ B. Ordered List <ol> — Numbered / Sequenced

Used when the ORDER of items matters (e.g., steps in a process, ranked items, instructions):

```
<!-- type attribute: "1"=numbers, "A"=uppercase, "a"=lowercase, "I"=Roman -->
<ol type="1" start="1">
  <li>Plan the website layout</li>
  <li>Write the HTML structure</li>
  <li>Apply CSS styling</li>
  <li>Add JavaScript interactivity</li>
  <li>Test on different browsers</li>
</ol>
```

▸ C. Description List <dl> — Term and Definition Pairs

Used for glossary-style lists with terms and their corresponding descriptions:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language – defines the structure of web pages.</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets – controls the visual appearance of HTML
elements.</dd>

  <dt>JavaScript</dt>
  <dd>A programming language that adds interactivity and dynamic behavior.</dd>
</dl>
```

6.2 Anchor Tags and Hyperlinks

The <a> (anchor) tag creates clickable hyperlinks that navigate users to other pages, sections, email addresses, or downloadable files:

```
<!-- 1. External link – opens in a new browser tab -->
```

```
<a href="https://www.google.com" target="_blank" title="Go to Google">Visit
Google</a>

<!-- 2. Internal link — goes to another page in same website -->
<a href="about.html">About Us</a>

<!-- 3. Page section bookmark — jumps to specific section on same page -->
<a href="#contact-section">Go to Contact Section</a>

<!-- 4. Email link — opens the user's email application -->
<a href="mailto:info@darshan.ac.in">Email Us</a>

<!-- 5. Phone link — opens dialer on mobile devices -->
<a href="tel:+911234567890">Call Us</a>

<!-- 6. Download link — forces file download instead of opening -->
<a href="files/syllabus.pdf" download>Download Syllabus PDF</a>
```

Attribute	Purpose	Example Value
href	Specifies the destination URL or resource	href="https://www.darshan.ac.in"
target	Controls where the link opens	_blank (new tab) | _self (same tab)
title	Tooltip text displayed on hover	title="Visit our homepage"
download	Forces the browser to download the file	download (no value needed)
rel	Relationship between pages (for SEO)	rel="noopener noreferrer"

6.3 Image Tags

The <img> element embeds images into the web page. It is a self-closing (void) element — it has no closing tag:

```
<!-- Basic image with all recommended attributes -->
<img
  src="images/campus-photo.jpg"
  alt="Darshan University Campus"
  width="600"
  height="400"
  loading="lazy"
  title="Our Beautiful Campus"
>

<!-- Clickable image link (image wrapped inside anchor tag) -->
<a href="gallery.html">
  <img src="images/thumbnail.jpg" alt="View Gallery" width="200">
</a>
```

Attribute	Purpose	Why It Matters
src	File path or URL of the image	Without it, the image cannot be found
alt	Descriptive text if image fails to load	Essential for accessibility & SEO. Screen readers use this.

width / height	Explicit size in pixels	Prevents page layout shift while image loads
loading="lazy"	Delays loading until user scrolls near it	Improves page load speed significantly
title	Tooltip text shown on hover	Provides extra context to the user

⚠️ **Always add alt text to every image:**

Never leave alt empty unless the image is purely decorative.

Good alt text: alt="Students studying in the computer lab"

Bad alt text: alt="image" or alt="img123.jpg"

7. HTML Tables & Forms

7.1 HTML Tables

HTML tables are used to display structured data in rows and columns — like a spreadsheet. Remember: tables are for DATA only, never for page layout (use CSS Flexbox or Grid for layout).

▸ Core Table Elements:

Tag	Full Name	Purpose
<table>	Table	Root container that wraps the entire table
<thead>	Table Head	Groups the header row(s) — displayed at top
<tbody>	Table Body	Groups the main data rows
<tfoot>	Table Foot	Groups footer/summary rows — displayed at bottom
<tr>	Table Row	Defines one horizontal row of cells
<th>	Table Header	Header cell — bold & centered by default
<td>	Table Data	Standard data cell — normal text
<caption>	Caption	Title text shown above the table

▸ Complete Table Example with colspan and rowspan:

```
<table border="1" cellpadding="8" cellspacing="0">
  <caption>Student Grade Report — Semester 1</caption>

  <thead>
    <tr>
      <th>Student ID</th>
      <th>Student Name</th>
      <th>Course</th>
      <th>Grade</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>101</td>
      <td>Rahul Patel</td>
      <td rowspan="2">Computer Science</td>  <!-- Merges 2 rows -->
      <td>A+</td>
    </tr>
    <tr>
      <td>102</td>
      <td>Priya Shah</td>
      <!-- rowspan cell continues here — no <td> needed -->
      <td>A</td>
    </tr>
    <tr>
      <td colspan="3">Total Enrolled Students</td>  <!-- Merges 3 columns -->
```

```
      <td><strong>2</strong></td>
    </tr>
  </tbody>
</table>
```

colspan vs rowspan:

colspan="3" → Merges the cell ACROSS 3 columns horizontally (left to right)

rowspan="2" → Merges the cell DOWN across 2 rows vertically (top to bottom)

7.2 HTML Forms

HTML Forms are interactive elements that collect input from users and submit it to a server for processing. They are used for login pages, registration, search bars, contact forms, and much more.

```
<form action="/process_form.php" method="POST" enctype="multipart/form-data">

  <!-- Text Input -->
  <label for="fullname">Full Name:</label>
  <input type="text" id="fullname" name="fullname" placeholder="e.g. Rahul Patel"
required>

  <!-- Password Input -->
  <label for="password">Password:</label>
  <input type="password" id="password" name="password" required>

  <!-- Email Input (has built-in format validation) -->
  <label for="email">Email Address:</label>
  <input type="email" id="email" name="email" required>

  <!-- Number Input with range -->
  <label for="age">Age:</label>
  <input type="number" id="age" name="age" min="17" max="30">

  <!-- Radio Buttons (user can select ONLY ONE) -->
  <label>Gender:</label>
  <input type="radio" name="gender" value="male" id="male">
  <label for="male">Male</label>
  <input type="radio" name="gender" value="female" id="female">
  <label for="female">Female</label>

  <!-- Checkboxes (user can select MULTIPLE) -->
  <label>Interests:</label>
  <input type="checkbox" name="interest" value="coding" id="coding">
  <label for="coding">Coding</label>
  <input type="checkbox" name="interest" value="design" id="design">
  <label for="design">Web Design</label>

  <!-- Dropdown Select -->
  <label for="branch">Branch:</label>
  <select id="branch" name="branch">
    <option value="">-- Select Branch --</option>
    <option value="cse">Computer Science (CSE)</option>
    <option value="it">Information Technology (IT)</option>
    <option value="ec">Electronics (EC)</option>
```

```
  </select>

  <!-- Multi-line Text Area -->
  <label for="address">Address:</label>
  <textarea id="address" name="address" rows="4" cols="50"
            placeholder="Enter your full address"></textarea>

  <!-- File Upload -->
  <label for="photo">Upload Photo:</label>
  <input type="file" id="photo" name="photo" accept=".jpg,.png">

  <!-- Date Picker -->
  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob" name="dob">

  <!-- Submit and Reset Buttons -->
  <button type="submit">Submit Form</button>
  <button type="reset">Clear All Fields</button>

</form>
```

▸ GET vs POST — Key Comparison:

Feature	GET Method	POST Method
Data Location	Appended to URL: ?name=Rahul&age=20	Hidden inside HTTP Request body
Security	Insecure — passwords visible in URL!	Secure — data not visible in address bar
Data Size Limit	~2048 characters (URL length limit)	No practical size limit (handles file uploads)
Bookmarkable?	Yes — URL can be saved as bookmark	No — cannot be bookmarked
Use For	Search queries, page navigation	Login forms, registration, file uploads
Browser History	Saved in browser history	NOT saved in browser history

8. Multimedia, HTML5 Semantic Elements & Validation

8.1 Multimedia Tags

HTML5 introduced native support for audio and video, eliminating the need for third-party plugins like Adobe Flash. These tags work directly in all modern browsers.

▸ A. Audio Element <audio>

```
<audio controls autoplay loop preload="auto">
  <!-- Provide multiple formats for browser compatibility -->
  <source src="media/lecture.mp3" type="audio/mpeg">
  <source src="media/lecture.ogg" type="audio/ogg">
  <!-- Fallback message for old browsers -->
  Your browser does not support the audio element.
</audio>
```

Attribute	Effect
controls	Shows the play/pause/volume/seek control bar to the user
autoplay	Automatically starts playing when the page loads (may be blocked by browser)
loop	Repeats the audio continuously after it ends
muted	Starts audio in muted state (required for autoplay in Chrome)
preload	auto = preload all | metadata = preload info only | none = don't preload

▸ B. Video Element <video>

```
<video width="720" height="405" controls poster="images/video-thumbnail.jpg">
  <source src="media/lecture.mp4" type="video/mp4">
  <source src="media/lecture.webm" type="video/webm">
  <track kind="subtitles" src="subtitles.vtt" srclang="en" label="English">
  Your browser does not support the video tag.
</video>
```

Supported Media Formats:

Audio: MP3 (most compatible), OGG, WAV

Video: MP4/H.264 (most compatible), WebM, OGG

Always provide 2 source formats using <source> tags for maximum browser support

poster attribute shows a thumbnail image before the video starts playing

▸ C. Inline Frame <iframe>

An iframe embeds an external webpage or resource inside your page — commonly used to embed YouTube videos, Google Maps, or social media widgets:

```
<!-- Embed a YouTube video -->
```

```
<iframe
  width="560" height="315"
  src="https://www.youtube.com/embed/VIDEO_ID"
  title="HTML Tutorial Video"
  allowfullscreen
  loading="lazy">
</iframe>

<!-- Embed Google Maps -->
<iframe
  src="https://maps.google.com/maps?q=Darshan+University&output=embed"
  width="600" height="400"
  style="border:0;"
  allowfullscreen
  loading="lazy">
</iframe>
```

8.2 HTML5 Semantic Layout Tags

Before HTML5, developers used meaningless , tags everywhere. HTML5 introduced semantic elements that describe the PURPOSE of content clearly — helping browsers, search engines, and screen readers understand your page structure.

Semantic vs Non-Semantic:
❌ Old Way (Non-Semantic): , ,
✅ New Way (Semantic HTML5): <header>, <nav>, <main>, <section>, <article>, <footer>
Semantic tags make code easier to read, maintain, and rank in search engines.

HTML5 Tag	Purpose	Typical Content
<header>	Page or section header	Logo, site title, main navigation links
<nav>	Navigation links block	Main menu, sidebar links, breadcrumbs
<main>	Primary page content (only ONE per page)	The main unique content of each page
<section>	Thematic section of content	Grouped topics: About, Services, Features
<article>	Self-contained, reusable content	Blog posts, news articles, forum replies
<aside>	Tangential sidebar content	Related links, ads, widgets, tips
<footer>	Page or section footer	Copyright, social links, privacy policy
<figure>	Media with caption	Images, diagrams, charts with <figcaption>

```
<!-- Complete HTML5 Semantic Page Structure Example -->
<!DOCTYPE html>
<html lang="en">
<head>
```

```
  <meta charset="UTF-8">
  <title>Darshan University – Web Designing</title>
</head>
<body>

  <header>
    <img src="logo.png" alt="University Logo">
    <h1>Darshan University</h1>
    <nav>
      <a href="index.html">Home</a>
      <a href="courses.html">Courses</a>
      <a href="contact.html">Contact</a>
    </nav>
  </header>

  <main>
    <section id="about">
      <h2>About Web Designing</h2>
      <p>Web designing is the process of creating websites...</p>
    </section>

    <article>
      <h2>Chapter 1: HTML Fundamentals</h2>
      <p>HTML is the building block of every website...</p>
    </article>

    <aside>
      <h3>Quick Links</h3>
      <ul>
        <li><a href="#">Download Notes</a></li>
        <li><a href="#">Past Papers</a></li>
      </ul>
    </aside>
  </main>

  <footer>
    <p>&copy; 2025 Darshan University. All rights reserved.</p>
  </footer>

</body>
</html>
```

8.3 Native HTML5 Form Validation

HTML5 provides powerful built-in validation attributes that check user input directly in the browser — no JavaScript needed for basic validation:

```
<form>

  <!-- Required field – cannot submit without filling this -->
  <input type="text" name="username" required placeholder="Enter username">

  <!-- Email format validation – checks for @ and domain -->
  <input type="email" name="email" required placeholder="you@example.com">

  <!-- Number range constraint – only 18 to 60 allowed -->
```

```
  <input type="number" name="age" min="18" max="60">

  <!-- Phone number — must be exactly 10 digits -->
  <input type="tel" name="phone" pattern="[0-9]{10}"
         placeholder="10-digit mobile number">

  <!-- URL format validation -->
  <input type="url" name="website" placeholder="https://your-website.com">

  <!-- Text length constraint -->
  <input type="text" name="bio" minlength="20" maxlength="200"
         placeholder="Write a short bio (20-200 chars)">

  <!-- Auto-focus on this field when page loads -->
  <input type="search" name="q" autofocus placeholder="Search...">

  <button type="submit">Submit</button>
</form>
```

▸ HTML5 Validation Attributes — Summary:

Attribute	What It Does	Example
required	Field cannot be empty — form won't submit without it	<input required>
pattern	Input must match a Regular Expression (Regex) pattern	pattern="[A-Z]{3}[0-9]{4}"
min / max	Minimum and maximum value for numbers or dates	min="18" max="60"
minlength / maxlength	Minimum/maximum number of characters allowed	minlength="8" maxlength="20"
type="email"	Automatically validates email format (must have @ and domain)	<input type="email">
type="url"	Validates URL format (must start with http:// or https://)	<input type="url">
type="number"	Only accepts numerical input	<input type="number">
autofocus	Automatically moves cursor to this field on page load	<input autofocus>
autocomplete	Suggests previously entered values from browser history	autocomplete="on"

Quick Reference — HTML Tags Summary

Use this page as a quick cheat sheet while writing HTML code:

▸ Structure Tags

Tag	Purpose
<!DOCTYPE html>	Declares HTML5 document
<html>	Root element
<head>	Metadata container (not displayed)
<body>	Visible page content
<title>	Browser tab title
<meta>	Page metadata
<link>	Links external CSS file
<script>	Links or embeds JavaScript

▸ Text & Formatting Tags

Tag	Purpose
<h1> to <h6>	Headings (h1 = largest, h6 = smallest)
<p>	Paragraph of text
 	Line break
<hr>	Horizontal dividing line
	Important bold text
	Emphasized italic text
<mark>	Highlighted text
<code>	Inline code snippet
<pre>	Preformatted text (preserves spaces)
<blockquote>	Indented quotation from external source
	Inline container (for styling text)
	Block-level container (for grouping sections)

▸ List Tags

Tag	Purpose

	Unordered (bullet) list
	Ordered (numbered) list
	List item (used inside ul or ol)
<dl>	Description/definition list
<dt>	Definition term
<dd>	Definition description

▸ Media Tags

Tag	Purpose
<a>	Hyperlink / anchor tag
<img>	Embeds an image
<audio>	Embeds audio player
<video>	Embeds video player
<source>	Specifies media file source (inside audio/video)
<iframe>	Embeds external webpage
<figure>	Container for media with caption
<figcaption>	Caption for figure element

▸ Table Tags

Tag	Purpose
<table>	Creates a table
<thead>	Table header section
<tbody>	Table body section
<tfoot>	Table footer section
<tr>	Table row
<th>	Table header cell (bold, centered)
<td>	Table data cell
<caption>	Table title

▸ Form Tags

Tag / Attribute	Purpose
<form>	Creates an interactive form

<input>	Single-line input field (many types)
<textarea>	Multi-line text input
<select>	Dropdown list
<option>	Individual dropdown item
<button>	Clickable button
<label>	Label text for an input field
type="text"	Plain text input
type="password"	Hidden/masked input
type="email"	Email input with validation
type="checkbox"	Multiple-choice tick box
type="radio"	Single-choice circular button
type="file"	File upload button
type="submit"	Submit the form data
required	Makes field mandatory

▸ HTML5 Semantic Tags

Tag	Purpose
<header>	Page/section header with logo and nav
<nav>	Navigation links block
<main>	Primary unique content of the page
<section>	Thematic content section
<article>	Independent reusable content
<aside>	Sidebar / tangential content
<footer>	Page/section footer

Practice Questions — Chapter 1

Instructions:

Attempt all questions. Short answer questions carry 2 marks each. Long answer questions carry 5 marks each.

For coding questions, write the complete HTML code with proper indentation and comments.

Section A — Short Answer Questions

1. What is the difference between the Internet and the World Wide Web?

2. What does HTTP stand for? Name any two HTTP request methods.

3. What is the role of the <head> and <body> tags in HTML?

4. What is the purpose of the alt attribute in the <img> tag?

5. Differentiate between <b> and tags with examples.

6. What is the difference between colspan and rowspan in HTML tables?

7. What is the difference between GET and POST methods in HTML forms?

8. Name any four HTML5 semantic tags and state their purpose.

9. What does the required attribute do in an HTML form?

10. What is the purpose of the <meta name="viewport"> tag?

Section B — Long Answer Questions

1. Explain Client-Server Architecture with a neat diagram. Describe the role of HTTP in web communication. (5 marks)

2. Write an HTML5 page with: a header with your name and navigation links, a main section with a paragraph and an ordered list of 5 web technologies, and a footer with copyright text. (5 marks)

3. Create an HTML table showing the marks of 4 students in 3 subjects. Use <thead>, <tbody>, <tfoot>, colspan, and rowspan attributes appropriately. (5 marks)

4. Design a student registration form that collects: Full Name (text, required), Email (email, required), Mobile Number (10 digits, pattern validation), Branch (dropdown with 4 options), Gender (radio buttons), Interests (checkboxes), Photo upload, and Submit/Reset buttons. (5 marks)

5. Explain any six HTML5 semantic tags with their purpose and a short code example for each. (5 marks)

6. What are meta tags? Explain the purpose of any five different meta tags used in HTML with proper syntax. (5 marks)

Section C — Coding Exercises

1. Write HTML code to create a page that shows: an audio player with controls, a video player with controls and a poster image, and an iframe embedding any website. (5 marks)

2. Create a nested HTML list showing a university → departments → subjects hierarchy using both ordered and unordered lists. (3 marks)

3. Write HTML code for an anchor tag that: opens an external link in a new tab, links to a section on the same page, and creates an email link. Explain each attribute used. (3 marks)